

Introductory Lecture

COMPSCI 4ZP6A

Capstone project

2026

Instructor

- Mehdi Moradi - moradm4@mcmaster.ca
 - Office: ABB-C534
- TAs:
 - Amirhossein Sabour - saboua4@mcmaster.ca
 - Sahib Khokhar - khokhs5@mcmaster.ca
 - Tom Overall - overallt@mcmaster.ca

About the instructor



- Mehdi Moradi, PhD, PEng
- PhD, Computing, 2008, Queen's
- Academic work 2008-2014 @UBC and @Harvard Medical School
- Industry roles at IBM, Google, Meta
 - Worked as senior staff data scientist, and tech lead for multiple projects
- Associate Professor at McMaster Computing and Software since 2023

What is this course about?

- **The mission of this course (formal version):** Students, in teams of three to six members, undertake a substantial project in an area of computer science by performing each step of the software life cycle.
- **The mission of this course (informal version):** To give you an opportunity to experience “building” a system or “solving a real-world problem” while also managing the messy realities of teamwork in an imperfect world. That is why we insist on teams of at least 5 students in size.
- The focus of the CS version of Capstone is on solving a problem.

Learning outcomes

- Knowledge of product planning process, including Software Requirement Specification (SRS), Project Plan and Design Documents.
- Learn to work as part of a software development team.
- Learn and implement a prioritization system for the features of your proposed project/product.
- Deliver on the requirements as defined in your Requirement Specification

Classes, do I need to attend?

- **Mandatory In-Person Lectures (1:30 PM – 2:20 PM):**
 - Week 1: Wednesday Sep 9 and Thursday Sep 10, 2026
 - Thursday Sep 24, 2026
- **Mandatory In-Person Presentation Events:**
- **Proof of Concept Presentation:**
 - Monday Nov 23, 2026 (3 hours total: 9:30 AM – 11:20 AM Location TBD; 1:30 PM – 2:20 PM in regular classroom).
- **Final Course Presentations:**
 - **Final Course Presentations:** Monday April 5, 2027 (3 hours total: 9:30 AM – 11:20 AM Location TBD; 1:30 PM – 2:20 PM in regular classroom).

Important:

- You can NOT join a team in another ENG capstone class
- Team sizes are strictly 5-7

Milestones and grades

Milestone / Deliverable	Deadline	% of Overall Grade
Team and project selection: • Form a team • Select a project (title and short description) • Create GitHub repo and add TA and instructor	Sep 21, 2026	5%
Potential revision of project descriptions/selections based on feedback for student-proposed projects (project topics frozen after this date)	Sep 28, 2026	0%
Unified Software Requirement Specification (SRS) and Project Plan documentation	Oct 9, 2026	15%
Proof of concept demonstration: Two-minute video clip. (Note: Tutorial and class time combined for a 3-hour session; location TBD for tutorial time)	Nov 23, 2026	10%
Design and V&V document	Jan 25, 2027	10%
Final demonstration: Two-minute video clip. (Note: Tutorial and class time combined for a 3-hour session; location TBD for tutorial time)	April 5, 2027	5%
Final submission: Source code repository + finalized design document + User guide (all on GitHub) + working app/website + Poster PDF	April 5, 2027	50%
EXPO Demonstration and poster	TBD	5%
Total	-	100%

Meetings TA and Instructors (mandatory)

- Apart from the dates reserved above, use the other class times to hold meetings and advance your work

- TA office hours: You need to have at least one deep dive with your TA each half. They keep track of this.

- Tom: Monday 1:3–2:30
- Sahib: Wed 1:30-2:30
- Amir: Thursday 1:30-2:30

Instructor office hour / support sessions: **Mondays 9:30-11:30 on General Channel of Teams**

- For the next two Mondays, treat this as an open session: join on General channel and ask questions. After teams are formed, I post a sign-up sheet and you book a time slot.
- Book at least twice each semester.

Communication

- Teams will be our official communication mechanism
 - I will primarily use 'General Channel'. Important files (including lecture slides) are stored under *General Channel -> Files*
 - When teams are formed, your TAs will create a channel for each team. We will use this space to send messages that are specific to your team and you can use them to meet.
 - Do not start meetings in the general channel. Leave that to the instructor.
- After groups are formed:
 - Please use your group's channel on Teams to ask questions. Emails are rarely needed.
 - If you have to use email, copy all group members and include your group number in subject line.

Microsoft Teams interface showing a channel named "COMPSCI 4ZP6A:Capstone..." with a "General" channel selected. The channel's "Files" tab is active, displaying a list of documents and folders. A red box highlights the "General" channel in the left sidebar. A red box highlights the "meet_instructor_appointment_sheet" file in the file list, with a red arrow pointing to it from a text box. Another red box highlights the "sample_work_previous_year" folder in the file list, with a red arrow pointing to it from a text box. A third red box highlights the "Class Materials" folder in the file list, with a red arrow pointing to it from a text box.

Activity
Chat
Teams
Assignme...
Calendar
Calls
OneDrive
Main Channels
General
Form Teams

CC
General
Posts
Files
Avenue
meet_instructor.xlsx

+ New
Upload
Edit in grid view
Share
All Documents

Documents > General

Name	Modified	Modified By	+ Add column
Class Materials		Mehdi Moradi	
lecture_slides	About an hour ago	Mehdi Moradi	
meet_instructor_appointment_sheet	4 minutes ago	Mehdi Moradi	
project_list	About an hour ago	Mehdi Moradi	
sample_work_previous_year	About an hour ago	Mehdi Moradi	
COURSE_OUTLINE_2024.pdf	4 days ago	Mehdi Moradi	

The spreadsheet to book time with instructor is in this folder (also tab at the top)

Example SRS, and other docs.

Administrative details (1)

- Teams will be our official communication mechanism. Please check often!
- All deliverables (except the source code) should be submitted on Avenue by 11:59 pm EST on the day of deadline.
- Only PDF files are accepted as submitted work on Avenue.
 - Exception: video files for proof of concept and final presentation
- Each team is required to have a GitHub
- Please maintain a copy of each deliverable submission on your GitHub repository, under a folder named “docs”. This version will become part of the final evaluation package. It can be revised during the year.

Administrative details (2)

- In your GitHub repository, create a folder for “meetings”. Each time you meet as a team, create a brief text document in this folder, listing the names of the attendees and the agenda of the meeting.
- Use your GitHub repo for code tracking and version control. Commit code under the folder “src”. The instructor and TAs will monitor code commits. If there is a dispute about lack of contributions by a team member, one way to verify the level of contributions is to investigate the code commits from each student and their meeting attendance.
- For each deliverable, we will post a template. Please follow.
- Team size: 5-7 students. These upper and lower limits will be strictly enforced.

Team composition

- Your team composition is up to you.
- I suggest you assign one person as program manager: this person will have the responsibility of planning the work.
- Take turns taking notes for meetings and posting on GitHub
- Make sure everyone (including the PM has technical contributions)

Forming a team (1)

- The first step is to form a team of five to seven students with broadly similar interests.
- If you are interested in working with a teammate from SE, or from outside of CAS, you will need to discuss your specific proposal with the course instructor. If you work on a SE project, you will be subject to requirements of engineering accreditation.
- As you will be working with your partners for two terms, select people you can get along with. Some questions you should ask of yourself and your future teammates are as follows:
- Will your teammates have adequate time to meet and fully participate in the project, including classroom activities and meetings with stakeholders?
- Are there enough gaps in everyone's schedules to hold meetings? Students on co-op terms may not be available for in-person meetings, or have restrictions on the hours they can work on the project.
- Do your teammates have the same, or at least compatible, work habits? When will you work together? All-nighters at the last minute? Regular afternoons? Make sure you pick people with a similar work style, or at least an approach that you can live with.

Forming a team (2)

- Do you want to produce a world-class deliverable, or just a good school project? There is no shame in producing a simple yet solid work, learning the principles of good project planning and execution at the same time. However, if your course load permits, this might be your opportunity to show the world — and future employers — what you can do. Either approach is acceptable, but be sure you and your colleagues share the same goals and aspirations, or you will all be frustrated.
- Does your potential teammates provide a good combination of talents and interests, so that collectively you can handle the different aspects of the project. Our projects will combine technical, interpersonal and English writing skills. You should make sure all of these skills are available within your team.
- Do your teammates share your interest in the theme/topic of your project?
- One note of caution: good friends do not always make good teammates! Look for people who complement your strengths and weaknesses, not just the people you know.

Selecting a project and “potential” supervisor (1)

- You can find a project from the list of project proposals (this is shared between SE and CS courses).
- Project list posted under files here (I also keep an offline copy on Teams, General Channel, under files, but that version is a snapshot and might not be the most recent list):

https://gitlab.cas.mcmaster.ca/courses/capstone/-/blob/main/ListOfPotentialProjects/PotentialProjects.pdf?ref_type=heads

- If you can't log in, you can reset your CAS password here: <https://idm.cas.mcmaster.ca> Click on the "Email" link to reset it via a token.

Selecting a project and “potential” supervisor (2)

- Projects can be carried out and supervised off-campus. Projects of this sort often arise naturally as the result of summer employment. To avoid complications, please be sure to discuss any industry collaborations with the instructor and the McMaster Industry Liaison Office (MILO) as soon as possible.

Selecting a project and “potential” supervisor (3)

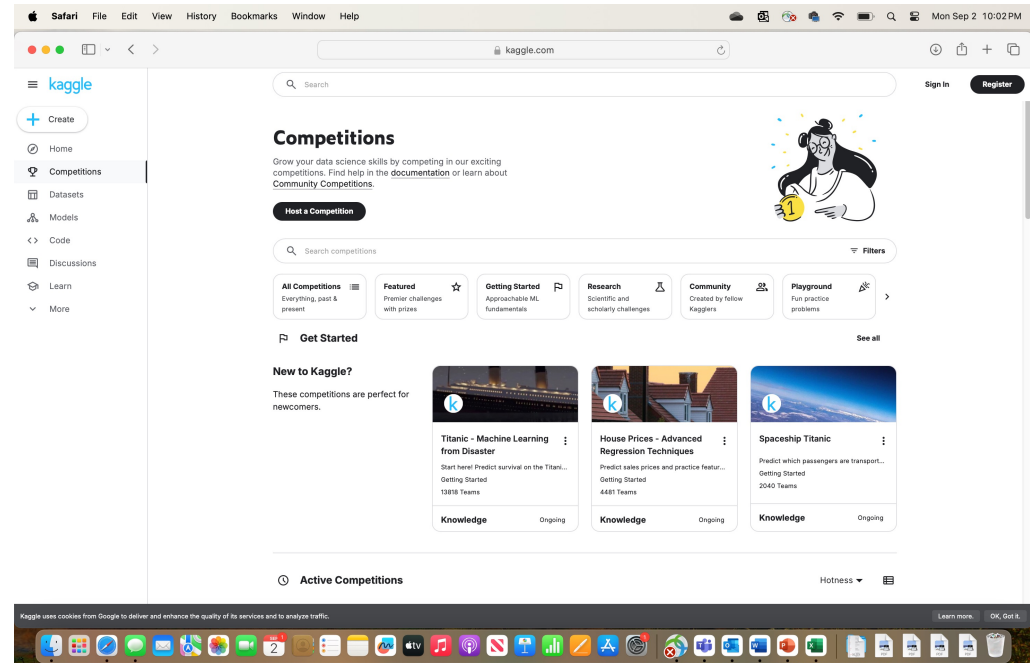
- In complex academic research projects, the emphasis in the evaluation process will be on finding a solution while maintaining the documentation requirements. But if the project has a conceptually simple solution, the students are expected to demonstrate excellence in collecting requirements, human computer interaction, testing and communicating with the supervisors.
- If you choose a project proposed by a faculty member or an external party, the proposer can choose to supervise the team on a technical level. But the grading will remain the responsibility of the instructor.

Choosing your own project

- Or you can propose your own project. Student proposed projects can come from interaction with CAS faculty members, outside-CAS faculty members, from industry, or from your own ideas. If relevant for your project, you should enlist the assistance of faculty, staff, or people from industry. A project may be an entirely original concept, continue an older topic, or repeat an older project, but with significant improvements.

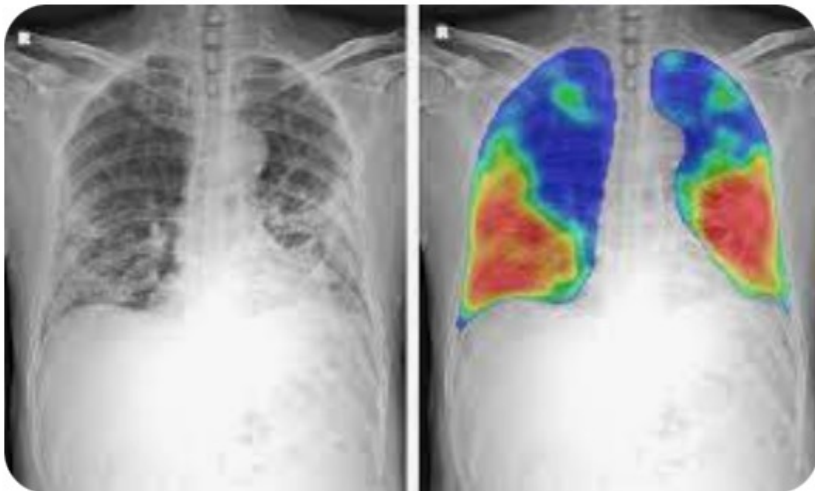
How to build your own project? Make a challenge into a project

- [Kaggle](#) competitions and datasets
- You can pair a tough problem with an app or website that performs the task that is the subject of the competition
- Examples of bad projects:
 - Building a website with content alone
 - Train an AI model
- Example of a good project
 - Build an AI model that solves a problem and develop an app or website to deliver that solution to users and build to high standards of design for user management and security



Example: Pneumonia challenge

- Train a model to detect the disease
- Deploy the model, allow user upload for disease detection
- Build user log-in and develop to high security standards



Featured Prediction Competition

RSNA Pneumonia Detection Challenge

Can you build an algorithm that automatically detects potential pneumonia cases?

\$30,000
Prize Money

RSNA Radiological Society of North America · 1,499 teams · 5 years ago

[Overview](#) [Data](#) [Code](#) [Discussion](#) [Leaderboard](#) [Rules](#) [Join Competition](#)

Overview

Start Aug 27, 2018 **Close** Oct 31, 2018

Merger & Entry

Description

In this competition, you're challenged to build an algorithm to detect a visual signal for pneumonia in medical images. Specifically, your algorithm needs to automatically locate lung opacities on chest radiographs.

Here's the backstory and why solving this problem matters.

Pneumonia accounts for over 15% of all deaths of children under 5 years old internationally. In 2015, 920,000 children under the age of 5 died from the disease. In the United States, pneumonia accounts for over 500,000 visits to emergency departments [1] and over 50,000 deaths in 2015 [2], keeping the ailment on the list of top 10 causes of death in the country.

While common, accurately diagnosing pneumonia is a tall order. It requires review of a chest radiograph (CXR) by highly trained specialists and confirmation through clinical history, vital signs and laboratory exams. Pneumonia usually manifests as an area or areas of increased opacity [3] on CXR. However, the diagnosis of pneumonia on CXR is complicated because of a number of other conditions in the lungs such as fluid overload (pulmonary edema), bleeding, volume loss (atelectasis or collapse), lung cancer, or post-radiation or surgical changes. Outside of the lungs, fluid in the pleural space (pleural effusion) also appears as increased opacity on CXR. When available, comparison of CXRs of the patient taken at different time points and correlation with clinical symptoms and history are helpful in making the diagnosis.

CXRs are the most commonly performed diagnostic imaging study. A number of factors such as positioning of the

Competition Host
Radiological Society of North America

Prizes & Awards
\$30,000
Awards Points & Medals

Participation
561 Competitors
1,499 Teams
2,001 Entries

Tags
[Image](#) [Medicine](#)

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For AI projects having data is mandatory

- Do not propose an AI or machine learning project without having data!

No stock market forecasting project!

- Please do not propose forecasting stock prices as a capstone project!